Project Abstract

Medical Insurance Cost Predictor

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The **Medical Insurance Cost Predictor** is a **Data Science** project designed to estimate an individual's **medical insurance cost** using health and demographic features. Factors such as **age, gender, BMI, number of children, smoking status, and region** can significantly influence insurance expenses. The main objective of this project is to develop a reliable prediction system that can estimate insurance charges based on these factors.

The project uses a **medical insurance dataset** containing historical records. The data is first subjected to **data cleaning and preprocessing** to handle missing values, duplicate records, and categorical features. **Exploratory Data Analysis (EDA)** is performed using statistical analysis and visualizations to identify important patterns and relationships between the features and insurance charges. **Regression algorithms**, such as **Linear Regression, Decision Tree Regression, and Random Forest Regression**, are then applied to build the prediction model. Model performance is evaluated using **MAE, MSE, RMSE.**

The proposed system provides an **estimated insurance cost** for an individual based on their input features. It helps identify the major factors affecting **medical expenses** and demonstrates the practical application of **Data Science and Machine Learning in healthcare analytics**. The prediction system can support **insurance companies in risk assessment** and help individuals gain a better understanding of their potential healthcare costs. Overall, the project provides a simple and effective approach for **insurance cost estimation and data-driven decision-making**.